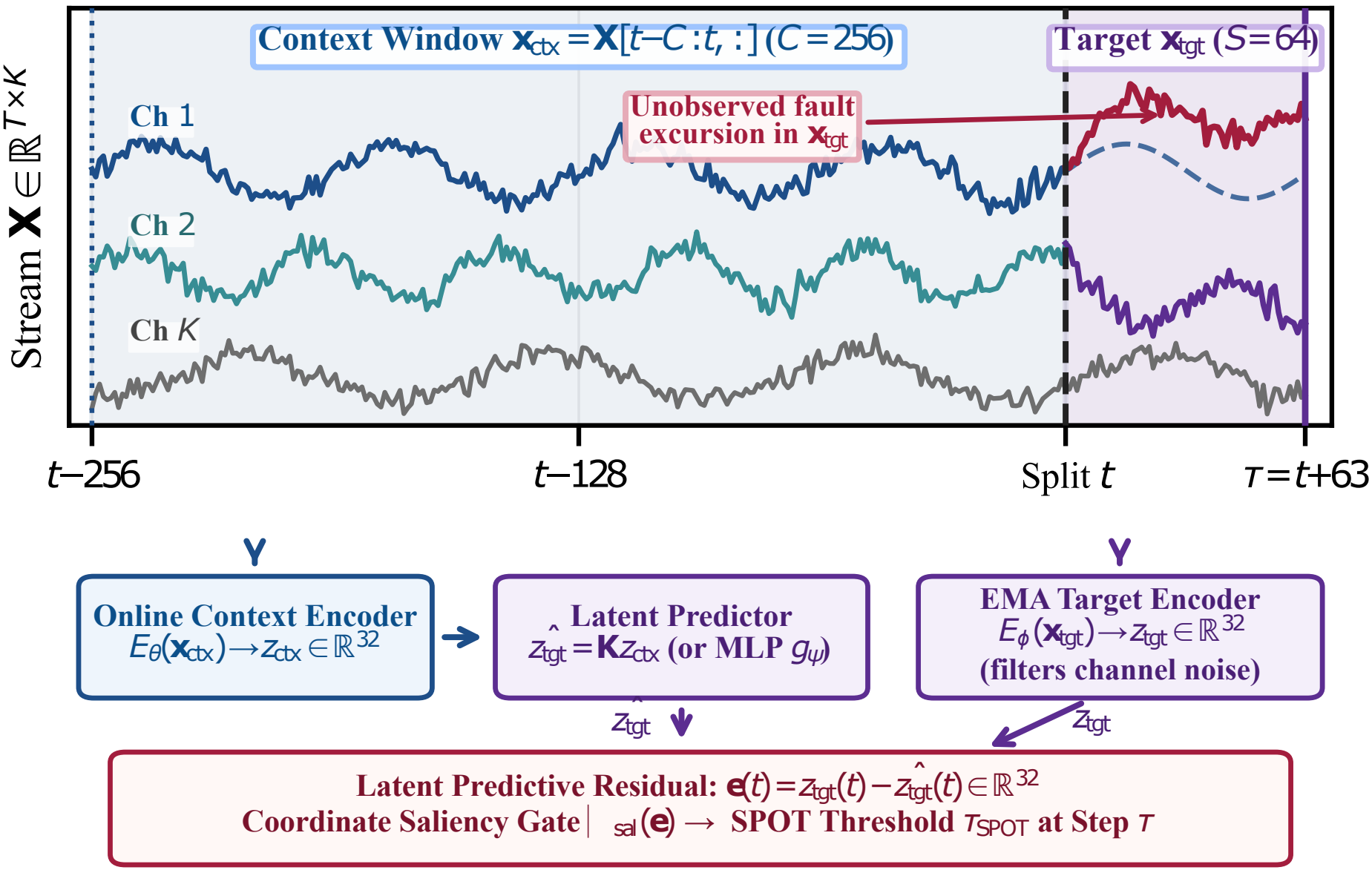

Sliding Window Partitioning ($W = C + S = 320$ steps, Stride 1)
Context Window $\mathbf{x}_{ctx} = \mathbf{X}[t-C:t, :]$ $(C = 256)$
Target $\mathbf{x}_{tgt}$ $(S = 64)$
Unobserved fault excursion in $\mathbf{x}_{tgt}$
Stream $\mathbf{X} \in \mathbb{R}^{T \times K}$
Ch 1
Ch 2
Ch K
$t-256$
$t-128$
Split t
$\tau = t+63$
Online Context Encoder
$E_\theta(\mathbf{x}_{ctx}) \to z_{ctx} \in \mathbb{R}^{32}$
Latent Predictor
$\hat{z}_{tgt} = \mathbf{K} z_{ctx}$ (or MLP g_ψ)
EMA Target Encoder
$E_\phi(\mathbf{x}_{tgt}) \to z_{tgt} \in \mathbb{R}^{32}$
(filters channel noise)
$\hat{z}_{tgt}$
z_{tgt}
Latent Predictive Residual: $\mathbf{e}(t) = z_{tgt}(t) - \hat{z}_{tgt}(t) \in \mathbb{R}^{32}$
Coordinate Saliency Gate $|_{sal}(\mathbf{e}) \to$ SPOT Threshold τ_{SPOT} at Step τ